

Aernest Lean Rivera

Senior Hardware Design Engineer

Address: Matungao Bulakan, Bulacan 3017

Phone: 09392672086

E-mail: aernestleanrivera@gmail.com

EDUCATION

2019-06 **Bachelor of Science in Electronics Engineering**
Bulacan State University - Malolos, Bulacan

PROFESSIONAL LICENSE

LICENSED ELECTRONICS ENGINEER (10/2019)
LICENSED ELECTRONICS TECHNICIAN (10/2019)

WORK HISTORY

August 2024 to July 2025 **Senior Hardware Design Engineer**
Packetworx Inc., Pasig City

- **Hardware System Design & Development:** Oversee end-to-end IoT hardware design and development, from component selection and schematic capture, PCB design to functional validation.
- **Prototyping & Performance Validation:** Manage device prototyping and ensure designs satisfy performance, reliability, and specification requirements.
- **Testing & Troubleshooting:** Create test plans and test procedures to solve complex hardware problems through systematic analysis and teamwork.
- **Research and Development Teamwork:** Closely interact with software, firmware, QA engineer and network teams to validate easy hardware integration.
- **Documentation & Traceability of Design:** Prepare complete documentation in the form of schematic design files, PCB design files, BOMs, Gerber and drill files, test procedures, and validation reports.
- **Review of Design & Compliance with Standards:** Give professional comments on schematics and PCB layouts for adherence to best practices and industry standards.
- **Cross-Functional Collaboration:** Work closely with marketing, product, and engineering teams to provide technical input for product brochures, datasheets, and user manuals, ensuring accurate specifications, clear technical messaging, and alignment with actual hardware performance

April 2021 to August 2024

Hardware Design Engineer

Packetworx Inc., Pasig City

- **Design and develop IoT hardware systems:** That consist of choosing viable circuit design, component selection, and schematic capture for creating innovative and effective solutions.
- **Prototyping and validation steps:** Conduct thorough prototyping processes to validate design principles, ensuring solid functionality and performance.
- **Develop and conduct thorough test plans:** Design effective test plans and procedures to ensure hardware performance, reliability, and conformance to specifications.
- **Troubleshoot and solve hardware problems:** Recognize and fix complex hardware issues using disciplined troubleshooting and analysis, working together with cross-functional teams to attain maximum solutions.
- **Document in-depth hardware design processes:** Develop detailed documentation for hardware designs, design reviews, test procedures, hardware bills of materials, and validation reports to provide traceability and knowledge transfer.
- **Work with multidisciplinary teams:** Share close coordination with application engineers, software developers, firmware engineers, network engineers, and other stakeholders to provide smooth integration of functional hardware parts.

March 2020 to October 2020

Associate Software Engineer

Accenture Technology

- Create software solutions, working alongside varied teams within an agile culture
- SAP ABAP, SAP HANA, and FIORI Bootcamp
- Japanese Language Training Bootcamp
- Service Desk Soft Skill Training Bootcamp.

TECHNICAL SKILLS

IoT Hardware Architecture & Circuit Design

- Advanced IC-level circuit analysis with strong expertise in interpreting datasheets, electrical characteristics, and design trade-offs for low-power and wireless systems
- Full product-cycle PCB development for IoT devices: component selection, schematic capture, multilayer PCB layout, BOM optimization, DRC, and Gerber generation for manufacturing

RF, Power & Signal Integrity (IoT-Focused)

- Design and debugging of low-power power supplies, battery-powered systems, and mixed-signal circuits
- Practical experience troubleshooting signal integrity, noise, grounding, and power stability issues in wireless and sensor-based devices

Hardware Validation & R&D Testing

- Extensive hands-on use of oscilloscopes, signal generators, programmable power supplies, Joulescope, and digital multimeters for characterization, power profiling, and performance validation
- Component-level fault isolation and rework using microscopes and professional soldering/rework stations

Embedded Systems & Firmware (IoT Platforms)

- Development of production test firmware and board-bring-up utilities for ESP32-based systems using Arduino IDE
- Proficient in C/C++ for embedded development and Python for test automation, data logging, and validation tools

Design, Simulation & Prototyping Tools

- KiCad, Altium Designer, Proteus, Multisim, Arduino IDE

PCBA & Manufacturing Enablement

- Prototype and production PCBA assembly, inspection, bring-up, and functional testing
- DFM/DFA collaboration with manufacturing teams to improve yield, reliability, and scalability of IoT products

LANGUAGES

Filipino

English

Basic Japanese

CERTIFICATIONS

The Things Certification - The Things Stack

The Things Certification- Advanced

The Things Certification - Fundamentals

The Things Certification - Network Management Semtech - Academy for LoRAWAN